

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

NMTC – 19TH October, 2024.

Sub junior Level - VII and VIII Grades – Final

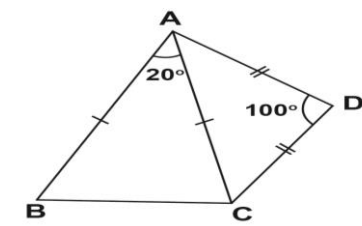
Instructions:

1. Answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your 'rough work' should be in the answer sheet itself.
6. Maximum time allowed is THREE hours.

1. ℓ, m, n are positive integers such that
 m divides $\ell+1$, n divides $m+1$ and ℓ divides $n+1$.
Find all possible triples (ℓ, m, n)
2. Consider 43 positive integers. Show that there exist two of them, say x and y , such that $x^2 - y^2$ is divisible by 100.
3. $ABCD$ is a quadrilateral and G, H are the middle points of AC and DB respectively. If CB, DA are produced to meet at E , prove that the area of $\triangle EGH$ is equal to $\frac{1}{4}$ (area of quadrilateral $ABCD$).
4. a, b, c are three non-negative real numbers whose sum is not greater than $\frac{1}{2}$. Show that
$$(1-a)(1-b)(1-c) \geq \frac{1}{2}$$
5. The vertical angle A of an isosceles triangle ABC is $\frac{1}{3}$ of each of the base angles. Two points M and N are taken on AB and AC respectively so that $BM = BC = CN$. If BN and CM cut at P , show that $\angle MPN = \angle B$.
6. Solve for x, y, z :
$$\begin{aligned} 12(x + y) &= 5xy \\ 24(y + z) &= 7yz \\ 8(z + x) &= 3zx \end{aligned}$$
7. Ten line segments are given. Each line segment has length between 1 cm and 55 cm. Prove that you can select three line segments such that they form a triangle.

8. ABC and DAC are two isosceles triangles, with $\angle BAC = 20^\circ$ and $\angle ADC = 100^\circ$.

Prove that $AB = BC + CD$.



End of Question Paper